ChatGPT

# Audit Report: Sylvie & Elias Educational Apps (June 2026)

## 1. Executive Summary

This audit evaluates four web-based applications designed for a family: **SylviePhonetics** (an early phonics game), **SylvieApp** (a creative play world), **Sleepy** (a bedtime video playlist), and **EliasApp2** (cause-and-effect play for a toddler). The goal is to determine whether they meet the developmental, educational and safety needs of Sylvie (4 years, 3 months) and Elias (2 years) as described in therapist reports, the family NDIS plan and the evidence-based design research. Each app was explored thoroughly across desktop, tablet and phone viewports, tested under slow connections and offline conditions, and compared against early-childhood science and Australian guidelines.

**Overall judgement.** The suite shows promise: it avoids ads and third-party tracking, uses appealing original art, invites co-play, and aligns with some EYLF outcomes. **SylviePhonetics** offers a variety of phonological-awareness games, **SylvieApp** provides a delightful world anchored in Sylvie's interests and includes calm-down activities, **Sleepy** aggregates bedtime media and breathing exercises, and **EliasApp2** uses vehicles and tools to invite parent narration. However, several elements reduce their educational and developmental value. Key concerns include lack of scaffolded feedback in SylviePhonetics; reward-style star collection in SylvieApp; Sleepy's reliance on long YouTube videos and bright screens near bedtime; and limited vocabulary modelling in EliasApp2. Some activities risk overstimulation, frustration or reinforcement of rigid perfectionism. Technical issues such as silent failures, missing alt text and inconsistent touch targets were found. Privacy risks arise from external YouTube links and unguarded navigation. Many recommendations are easily addressable within the next development cycle.

**Top strengths.** The apps are free of ads and monetisation, use original assets tied to the children's interests, and include explicit parent co-play prompts. They provide large tap targets and simple navigation, and they embed some co-regulation strategies (e.g., breathing page in SylvieApp). EliasApp2's emphasis on cause-and-effect with labels supports vocabulary development. The research report informs session lengths, co-play and phonics sequencing.

**Major risks.** Sleepy's long video links and use of copyrighted Disney clips conflict with paediatric sleep science and guidelines advocating screen-free periods 45–60 minutes before sleep. SylviePhonetics gives minimal guidance after mistakes and sometimes fails to respond, which can frustrate a child who struggles with perfectionism and emotional regulation. SylvieApp's star-collection unlocking system introduces extrinsic reward loops contrary to neurodiversity-affirming practice. EliasApp2 sometimes presents too many interactive elements per screen for a two-year-old and lacks consistent language modelling. None of the apps implement session timers or natural closing rituals as recommended by research.

**Top five changes.** (1) Add scaffolded hints and calm error feedback in SylviePhonetics; (2) remove or redesign star-collection and ensure choices remain limited and meaningful in SylvieApp; (3) replace Sleepy's external YouTube links with offline audio stories and enforce a screen-free handoff; (4) include parent-guided narration and vocabulary prompts in EliasApp2; and (5) implement natural closing rituals with session timers across all apps. With these changes, the suite would move from “usable with supervision” to “ready for regular family use.”

## 2. Child Needs and Learning Matrix

The following matrix summarises key needs, strengths and goals extracted from therapist reports, the family plan and research. Only concise phrases are used inside the table; explanatory prose follows.

| Child need, strength or goal | Source (doc & section) | What an app could support | What an app should avoid | Relevant apps |
|---|---|---|---|---|
| **Emotional regulation** | Sylvie family report (p. 7 ‘Two Brains’); OT plan (Emotional regulation strategies) | Calm error feedback; breathing prompts; characters modelling frustration and repair; parent co-play prompts | Shaming errors; punishment; time pressure; variable-ratio rewards | All apps |
| **Interoception & body signals** | Play therapy letter (goal focus) | Breath-awareness activities; “check-in” prompts before tasks; descriptive language about body sensations | Ignoring signals; encouraging screen time when tired or dysregulated | Sleepy, SylvieApp |
| **Frustration tolerance & flexible thinking** | OT handover (perfectionism, mistakes) | Scaffolded hints; celebrate persistence; offer choices; allow repetition without penalty | “Wrong” buzzer or immediate negative feedback; forced progression | SylviePhonetics, SylvieApp |

| Child need, strength or goal | Source (doc & section) | What an app could support | What an app should avoid | Relevant apps |
|---|---|---|---|---|
| **Communication of needs & feelings** | OT plan (language & communication goals) | Vocabulary labelling; opportunities to choose; emotion naming in stories; encourage telling a caregiver | Confusing instructions; ambiguous icons; heavy text requiring reading | SylvieApp, EliasApp2 |
| **Social cues & shared attention** | NDIS notes (declarative language and co-play) | Co-play prompts; show-and-tell moments; turn-taking mechanics; parent gates for new modules | Solo play loops; reward for ignoring adults | All apps |
| **Predictability & transitions** | OT plan (visual supports, routines) | Visual schedules; countdowns; “next step” prompts; natural closing rituals | Abrupt endings; autoplay next games; endless videos | Sleepy, SylviePhonetics |
| **Sensory needs & movement** | OT handover (movement seeking, sound sensitivity) | Break prompts encouraging stretching; audio off toggles; minimal sudden sounds; warm colour palettes | Loud sounds; flashing animations; non-skip intros; busy screens | All apps |
| **Child-led play & creativity** | OT plan (child-led play) | Open-ended art and construction; limited rules; diverse choices reflecting interests | Drill-like tasks; linear progression; heavy scoring | SylvieApp |

| Child need, strength or goal | Source (doc & section) | What an app could support | What an app should avoid | Relevant apps |
|---|---|---|---|---|
| **Co-regulation & attachment** | Play therapy letter (co-regulation reliance); research on co-regulation | "Call Dad over" prompts; joint achievements; no solo streaks; breathing together | Gamification that rewards solo use; emphasising independence | All apps |
| **Sleep routine support** | Research (Sleep & bedtime section) | Warm colours; audio-first stories; timed closing ritual; fade-to-black mode; no screen 45–60 min before sleep | Bright screens; long videos; algorithmic recommendations; external ads | Sleepy |
| **Phonological awareness & early phonics** | Research (phonics sequence); NSW curriculum section | Sequence from rhyme to blending to grapheme–phoneme mapping; pure Australian phoneme audio; decodable words; scaffolded hints | Mixed sight-word and phonics tasks; schwa-added phonemes; guessing from pictures; long sessions | SylviePhonetics |
| **Elias's vocabulary & cause-and-effect learning** | Toddler research (cause-and-effect & video deficit) | Tap-and-see reactions; label vehicles, tools and actions; parent narrates; limit interactive elements per screen to ≤3 | Passive videos; too many buttons; inconsistent vocabulary; reward loops | EliasApp2 |

| Child need, strength or goal | Source (doc & section) | What an app could support | What an app should avoid | Relevant apps |
|---|---|---|---|---|
| **Real-world play transfer** | NDIS notes (limit choices & emphasise real play) | Prompts to find objects in the room, go outside, or show a toy; closing rituals that send the child off screen | Endless digital tasks; encouraging staying on device | All apps |

**Discussion.** These needs emphasise the importance of calm, contingent feedback, predictable structure, co-regulation and real-world transfer. Both children have strong interests and cognitive strengths but need support with emotional regulation, transitions and flexible thinking. The apps should scaffold their learning and gently invite them back to offline play. Unmet needs reveal why some features (e.g., star collection) may be problematic.

## 3. Cross-App Scorecard

Scores are from 0 (poor) to 5 (excellent) and reflect current implementation relative to evidence. Brief explanations follow the table.

| Category | SylviePhonetics | SylvieApp | Sleepy | EliasApp2 |
|---|---:|---:|---:|---:|
| **Fun & child appeal** | 4 | 5 | 3 | 4 |
| **Developmental appropriateness** | 3 | 4 | 2 | 3 |
| **Educational value** | 3 | 3 | 1 | 2 |
| **Alignment with child needs** | 3 | 4 | 1 | 3 |
| **Neurodiversity-affirming design** | 2 | 3 | 1 | 3 |
| **Parent co-play** | 3 | 4 | 2 | 3 |
| **Real-world transfer** | 2 | 3 | 1 | 3 |
| **Child-safe UX** | 3 | 4 | 2 | 3 |
| **Accessibility** | 3 | 4 | 2 | 3 |
| **Privacy & safety** | 4 | 4 | 1 | 4 |
| **Technical reliability** | 3 | 4 | 2 | 3 |
| **Overall score** | **2.9** | **3.9** | **1.7** | **3.1** |

**Explanation. SylviePhonetics** is engaging and covers multiple phonological tasks, but inconsistent feedback and occasional silent errors limit educational value and risk frustration. **SylvieApp** delights with imaginative play and fosters emotional language; however the star-collection unlocking can shift attention to rewards. **Sleepy** is the weakest: its reliance on YouTube, bright screens and lengthy videos conflicts with paediatric sleep science and introduces copyright risks; yet its breathing page is a positive element. **EliasApp2** offers accessible cause-and-effect play and invites parent co-play, though vocabulary prompts and closing rituals need reinforcement.

# 4. Per-App Audit

## 4.1 SylviePhonetics

**Intended purpose.** To build phonological awareness and early phonics skills through short games for preschoolers, preparing Sylvie for NSW Kindergarten literacy. Activities include rhyme matching, syllable clapping, initial-sound selection, blending games and grapheme–phoneme tasks. A settings panel allows toggling gentle mode and parent assist.

**Strengths.** *Variety of games* and large tap targets keep engagement high. *Parent assist* mode invites adult guidance, and *gentle level* reduces difficulty. The art uses neutral colours and avoids overstimulation. Audio feedback is calm when correct and uses an Australian accent. The app's organisation follows a rough phonics progression from syllables to phoneme identification, supporting evidence-based learning sequences.

**Direct supports.** Supports early phonological skills; fosters active involvement through tapping and choice; encourages persistence by allowing repeated attempts; uses Sylvie's name in some examples for meaningful connection; has parent prompts in settings. Aligns partially with EYLF Outcome 5 (communication) by developing sound awareness.

**Indirect supports.** Without scaffolded hints, the child may still learn by trial and error; however, learning is slower. Some games incorporate categories like animals or objects that may interest Sylvie (fairies, farms) but not deeply personalised.

**Missing supports.** There is no clear session timer or closing ritual; games autoplay onto the next without a natural break, contrary to recommended session design. Hints are absent: wrong answers either produce a generic "no" audio or remain silent; there are no prompts encouraging listening again, segmenting the word or modelling the phoneme, which could teach frustration tolerance. There are no movement or breathing breaks despite Sylvie's need for sensory regulation. The app lacks co-regulation prompts inside the activities.

**Functional bugs.** During testing on Chrome for Android and iOS viewports, selecting an incorrect answer sometimes caused the next question to load without feedback. Rapid tapping occasionally froze the page requiring a refresh. The `Clap Syllables` game sometimes mis-counted taps when the network was slow.

**Developmental/educational gaps.** Many tasks rely on picking the correct letter from two options; once the letter's shape appears, the child may guess from the first letter or picture rather than processing the sound, undermining sound-to-letter mapping. Mixing letters and pictures in early tasks can blur whether the goal is phonological awareness or letter recognition. There is no bridging to decodable words or print in context. The generic praise ("Great job!") after correct answers does not provide contingent feedback or encourage reflection. There is no progress indicator; children may over-work or stop abruptly. Games do not model mistakes as learning opportunities, which could accentuate Sylvie's perfectionism.

**UX & accessibility issues.** Some interactive elements are small on mobile devices (<44 pt). The contrast between text and background is occasionally low in the `Word Wheel` game. Audio icons lack descriptive alt text for screen readers. There is no option for audio-only or captions for hearing-impaired caregivers. The keyboard mode (toggling for physical keys) is seldom needed for this age and may confuse children.

**Safety & copyright.** The app itself is self-contained and uses original art; there are no external links or ads. However, if wrong answers link to external resources in future, parent gates would be needed. All audio recordings should remain stored locally to avoid external streaming.

**Recommendations & acceptance criteria.**

- Add three-step scaffolded feedback for wrong answers: (a) gentle hint ("Listen to the first sound again"), (b) model the target phoneme without adding a schwa, (c) invite the child to try again. Acceptance: every incorrect response triggers at least one specific hint before the answer is modelled.

- Introduce session timers with natural closing rituals (e.g., after five rounds a character waves goodbye and prompts the child to show a parent). Acceptance: after 10–12 minutes of play the app transitions to a calming end screen and pauses automatically.

- Insert brief movement or breathing breaks between games ("Stand up and stomp like a dinosaur!") to support sensory regulation. Acceptance: every 5–7 minutes an interstitial screen appears inviting a movement break.

- Clearly separate phonological tasks (sounds, rhymes) from grapheme tasks (letters) until the child has mastered phoneme isolation, aligning with the recommended sequence. Acceptance: each mini-game targets a single stage; rhyme games never display letters.

- Add parent co-play prompts within games ("Ask Dad to say the word with you") to encourage joint attention. Acceptance: each game includes at least one natural prompt inviting caregiver involvement.

## 4.2 SylvieApp

**Intended purpose.** To provide a child-led imaginative world where Sylvie can explore her interests (fairies, farms, princesses) through play, creativity and storytelling. The app aims to support emotional regulation and language development through open-ended scenes and calm-down activities.

**Strengths.** The world map and scene art are beautiful and cohesive, capturing Sylvie's interests. Scenes like **Fairy Garden**, **Farm**, **Princess Room**, **Trampoline** and **Wand Maker** offer interactive items and encourage imagination. Each scene has limited interactive elements (4–6) appropriate for a four-year-old. The app uses large tap targets and simple navigation. A **Calm Down** page guides the child through belly breathing with supportive phrases ("You are safe," "God made you and loves you"). Parent prompts appear via a "Parent Mode" overlay, reminding adults to co-play and linking scenes to real-world activities ("Go outside and collect leaves"). The art and music avoid overstimulation. The **Ideas** count emphasises creativity rather than points.

**Direct supports.** The app fosters creativity and narrative expression, aligning with child-led play principles. Scenes embed emotional language ("Bunny looks frustrated") and choices that help Sylvie practise flexible thinking. The Calm Down page integrates interoceptive awareness and breathing regulation. Parent co-play prompts encourage sharing with a caregiver, fulfilling co-regulation goals. The spiritual phrasing may resonate with Sylvie's family values.

**Indirect supports.** Collecting "ideas" to unlock new features may motivate persistence. The game occasionally uses logic puzzles (sorting objects) that indirectly support executive function and categorisation. Scenes featuring movement (Trampoline, Swing) encourage the child to get up and jump, supporting sensory needs. There is an optional **Story Builder** that allows creating simple narratives by dragging characters and props.

**Missing supports.** There is no session timer or closing ritual; the world is open-ended and can be used indefinitely. The star-collection unlocking mechanism, even renamed "ideas," is extrinsic and risks fostering compliance rather than intrinsic enjoyment; Sylvie already struggles with perfectionism and may fixate on unlocking everything. Emotional language appears mainly in the Calm Down page; scenes rarely model frustration or repair in play narratives. The app could better embed social language (e.g., characters ask for help) and cues to check in with a caregiver when things go wrong. There is limited integration of Sylvie's home vocabulary or experiences (e.g., referencing her siblings or school). The parent prompts are hidden behind a toggle that might be missed.

**Functional bugs.** On tablets the world map occasionally scrolls horizontally when it should be static, causing misalignment with scene icons. Some items require precise tapping (e.g., small fairy wings) and may not register, especially on smaller screens. Rapid collection of ideas sometimes increments counters incorrectly or resets unlocked items after a page refresh.

**Developmental/educational gaps.** The open-ended nature is appropriate but there is little scaffolding to encourage language expansion; the app could prompt Sylvie to describe what she created or to retell a story to a parent. There is no bridging to literacy or numeracy (e.g., counting objects, naming letters). The reward system could inadvertently create pressure to keep playing. Emotional regulation strategies are contained in a separate Calm Down page rather than integrated into play narratives. The content does not explicitly incorporate the EYLF outcomes on community or identity beyond personalised art.

**UX & accessibility issues.** Text labels are small in some scenes; there is no audio narration for icons, which could support preliterate users. Colour contrast is generally adequate but some pastel objects (e.g., yellow text on pale background) are low contrast. There is no dark mode or accessibility mode for high-contrast or simplified visuals.

**Safety & copyright.** The app uses original content, with no ads or external links, which is excellent. The spiritual content should remain optional and within the family's values but avoid preaching to avoid alienating other users. If the app is shared publicly, clear copyright notices are needed.

**Recommendations & acceptance criteria.**

- Remove star/idea collection unlocking or redesign it to emphasise exploration rather than accumulation. For example, new features could unlock through collaborative achievements ("You and Mum built a castle; a fairy arrives!"). Acceptance: unlocking triggers are relational or narrative, not numerically driven.

- Integrate emotion modelling within scenes. Characters can express frustration, ask for help and show repair ("The pig knocked over her blocks; let's help rebuild"). Acceptance: at least one scene includes a mini-story where a character models naming an emotion and a solution.

- Add a session timer and natural closing ritual. After 10–12 minutes a gentle character invites Sylvie to show her creation to a caregiver and guides her to an offline activity (e.g., "Let's go make a real fairy garden outside"). Acceptance: the app automatically offers an end-of-play sequence after the timer.

- Include contextual parent prompts within the main interface rather than only in a toggle. Acceptance: each scene contains a visible prompt encouraging the caregiver to join or extend play.

- Provide optional audio narration for on-screen text and icons to support preliterate users and accessibility. Acceptance: tapping a label triggers an Australian-accented audio reading.

### 4.3 Sleepy

**Intended purpose.** To act as a calming bedtime companion providing breathing exercises, lullabies and bedtime stories, curated by the family. It should help Sylvie transition to sleep and reduce bedtime struggles.

**Strengths.** The site offers categories of bedtime content (e.g., hymns, lullabies, Bible stories) and includes a custom **Breathe with Sylvie** sequence with slow animation and soothing instructions. There is a "Sleepy mode" button and a "Dim lights" toggle which reduce brightness and show a warm amber filter. The text emphasises that the app should be used 45–60 minutes before lights out and not as the final step, reflecting research guidance. Parent instructions remind caregivers to watch with the child and to follow a consistent bedtime routine.

**Direct supports.** The breathing exercise addresses interoceptive awareness and calm. The "Dim lights" function shifts colour temperature to warmer tones, aligning with recommendations for bedtime screens. The variety of lullaby styles allows Sylvie to select music meaningful to her, supporting meaningful experiences. The introduction page mentions that the shelf is curated, emphasising parent control.

**Indirect supports.** The presence of Christian music may provide comfort aligned with the family's values. Some playlists have gentle nature visuals that could invite conversation.

**Missing supports.** Most videos are embedded from YouTube and range from 5 minutes to 12 hours. Many feature copyrighted Disney characters (Moana, Aladdin, Frozen) and film clips. This exposes Sylvie to algorithmic recommendations, ads (if not using premium), and potential comments. Long "sleep music" videos encourage leaving the screen on all night, directly conflicting with research which calls for screen removal 45–60 minutes before sleep and recommends audio-first content. There is no screen lock to prevent tapping or accidentally opening a new link. There is no session timer or automatic fade to audio-only mode. The "Pick for me" random button may present highly stimulating or irrelevant content. There is no integration of co-regulation prompts or transitions to offline books. For children under two, any screen time at bedtime is discouraged; yet the app has no mechanism to restrict use when Elias is present.

**Functional bugs.** On some mobile devices the "Dim lights" button flickers and does not persist across page navigations. Videos sometimes start with loud ads or default to high brightness until the user taps the dimmer. When offline, the site displays broken video thumbnails without fallback content. The "Sleepy mode" toggle sometimes fails to darken the page if the OS dark mode is disabled.

**Developmental/educational gaps.** Sleepy provides no active learning; it is essentially a video library. The reliance on passive viewing contradicts the Four Pillars of Learning — there is no active involvement or social interaction. Without controlling video length and brightness, the app can delay sleep onset by suppressing melatonin production. Some content (Disney movies) may be arousing, contain conflict or fast pacing and could trigger Sylvie's fear or frustration. There is little emphasis on real-world bedtime routines such as brushing teeth, putting on pyjamas or reading a physical book.

**UX & accessibility issues.** Tap targets for video thumbnails are small on mobile. The site lacks a true dark mode for visually sensitive users. Audio descriptions or captions are not available, and there is no text alternative summarising each video. Navigation relies heavily on scroll menus that can be overwhelming. Some external links open in new tabs without warning.

**Safety & copyright.** Embedding copyrighted Disney clips and film songs poses legal and ethical concerns. YouTube's algorithmic recommendations introduce privacy and safety risks (ads, comments, content not appropriate for a four-year-old). The site does not block video recommendations, and there is no parent gate when leaving the curated list. For children like Sylvie and especially Elias, who are sensitive to sudden noises and bright visuals, these exposures are problematic.

**Recommendations & acceptance criteria.**

- Replace YouTube embeds with locally hosted or licensed audio-only stories and music. Provide a curated library of short (3–5 minute) lullabies and Bible stories with no external links. Acceptance: no external YouTube URLs present; all media hosted within the app and playable offline.

- Enforce a "lights out" mode where the screen fades to black after the introduction and only audio continues. Acceptance: when the Sleepy sequence starts, the display dims to <20 % brightness and fades to black within 30 seconds unless the user taps to keep the screen active.

- Remove or clearly limit video lengths to under 30 minutes and restrict to audio-dominant content. Acceptance: the longest audio story is <30 minutes, and no loops play beyond the end; a message invites the parent to close the screen.

- Add a session timer and a clear closing ritual ("Let's turn off the device and read a real book"). Acceptance: after 10 minutes of content the app automatically prompts the caregiver to transition the child to offline bedtime routines.

- Provide gentle reminders that screens should not be used 45–60 minutes before sleep and discourage use for children under 2. Acceptance: a notice appears when the app is launched during the recommended screen-free period or when the profile age is under 2.

- Implement a parent gate before playing any media; require answering a simple math question or holding a button for 3 seconds to confirm the caregiver is present. Acceptance: children cannot start new media without parent confirmation.

## 4.4 EliasApp2

**Intended purpose.** To engage Elias, a two-year-old, in cause-and-effect play focusing on vehicles, tools and construction. Each activity is designed for short sessions and invites parents to narrate actions and label objects.

**Strengths.** The home page shows categories such as **Garbage Trucks**, **Construction**, **Cars & Trucks**, **Tools** and **Screwdriver World**. Each category contains simple interactions: tapping a truck to dump rubbish, dragging a digger to scoop soil, turning a screw with a screwdriver. The app uses large buttons, warm colours and limited interactive elements (usually 2–4 per screen). A prominent overlay at launch invites an adult to sit with Elias and narrate. Many scenes include a text label and a consistent icon ("Tap to dump") that appears across categories. There is a sound toggle that turns off effects.

**Direct supports.** The app encourages active involvement and immediate feedback: a tap makes the truck beep and dump trash, providing clear cause-and-effect. Vehicles and tools are familiar to Elias and meaningful to his interests. Parent prompts emphasise co-play and limit sessions to 10 minutes, aligning with research recommendations. Some scenes encourage movement ("Stand up and pretend to lift the garbage bin"), facilitating real-world transfer.

**Indirect supports.** Completing tasks may build fine-motor skill practise (dragging the screwdriver). The variety of vehicles broadens vocabulary when accompanied by parent narration. Scenes like "Build a road" may nurture spatial awareness.

**Missing supports.** The app seldom labels objects aloud; text appears on screen, but there is no audio narration or vocabulary modelling. Without a caregiver, Elias may silently watch tapping animations without learning new words, which weakens the social interaction pillar. Some screens have more than four tap points (e.g., Construction site with five different actions), exceeding recommended limits for toddlers and potentially causing cognitive overload. There is no session timer or closing ritual; the overlay recommending 10 minutes is easily dismissed. Some scenes encourage turning screws or driving vehicles repeatedly without connecting to offline play (e.g., building a real road with blocks). There is no parent gate to prevent unsupervised use.

**Functional bugs.** On low-end devices, dragging objects occasionally stutters. Some interactive elements remain visible after being tapped (e.g., the bin resets to full without animation). The sound toggle resets after page refresh. Long pressing on buttons inadvertently triggers text selection, interfering with gameplay.

**Developmental/educational gaps.** Without explicit vocabulary prompts, the app relies heavily on the parent to supply language. The absence of new verbs or adjectives limits learning. There is no measure of progress or encouragement of turn-taking with a caregiver. Scenes lack variation in challenge; children may quickly get bored or, conversely, become fixated on repetitive tapping without context. There is no bridging to real-world activities beyond the initial overlay.

**UX & accessibility issues.** Some screens require dragging diagonally, which can be challenging for a toddler's fine motor skills. Tap targets are appropriate but sometimes overlap, causing accidental activations. There is no audio description for visually impaired caregivers, and text instructions are small. Colour contrast is mostly good but could be improved in the Tools scene.

**Safety & copyright.** The app uses original art and sound effects; no external links or ads are present. Tools are depicted as safe toys; the Screwdriver scene includes a warning "Real electric tools are for grown-ups" which is positive. However, there is no gate to prevent a child from quickly switching to other categories without a caregiver. Given Elias's age, any unsupervised screen use is discouraged.

**Recommendations & acceptance criteria.**

- Add audio vocabulary prompts. When the child taps or drags an item, a friendly voice names the object and action ("Dump the rubbish into the truck," "The excavator is digging"). Acceptance: each interactive element triggers a concise Australian-accented label when audio is on.
- Limit interactive elements per screen to three and emphasise depth over breadth (e.g., in Construction, allow only building a road, stacking blocks and using a crane). Acceptance: no screen contains more than three primary tap or drag interactions; any additional actions are unlocked sequentially or hidden.
- Implement a session timer with a clear closing ritual: after 5–8 minutes the app should prompt "Time to play with your real trucks!" and transition the child off screen. Acceptance: once the timer expires, interactive controls disable and a friendly character invites offline play.
- Enhance parent prompts: include on-screen text like "Try making the truck sound together" or "Ask Dad what this tool is called" to encourage co-play. Acceptance: each scene has at least one visible prompt for caregiver narration.
- Provide a parent gate to start or switch categories (e.g., hold for 3 seconds), ensuring adults control transitions. Acceptance: children cannot change categories without a long press or parent code.

## 5. Functional QA Bug Log

| ID | App | Screen/ feature | Steps to reproduce | Expected result | Actual result | Severity | Recommended fix |
|---|---|---|---|---|---|---|---|
| **BUG-1** | SylviePhonetics | Incorrect answer handling | In "First Sound Garden", select the wrong letter twice quickly | Child receives a hint, sees which sound was mis-identified | The next question loads without any feedback; sometimes silent [1] | P1 | Implement scaffolded hints and delay progression after incorrect answers |
| **BUG-2** | SylviePhonetics | Clap Syllables timing | In poor network conditions, tap quickly on the clap button | Taps are counted accurately; audio and animation stay synced | Taps are mis-counted; audio lags and resets unexpectedly | P2 | Debounce taps and handle offline mode; preload audio |

| ID | App | Screen/ feature | Steps to reproduce | Expected result | Actual result | Severity | Recommended fix |
|---|---|---|---|---|---|---|---|
| **BUG-3** | SylviePhonetics | Rapid tapping freeze | Tap letters repeatedly in any game for 10 seconds | Game continues responding | Game freezes; refresh needed | P2 | Throttle input; show loading indicator |
| **BUG-4** | SylvieApp | World map scrolling | On tablet, drag the world map horizontally | Map remains static or snaps back | Map scrolls and misaligns icons | P2 | Disable horizontal scroll or lock map position |
| **BUG-5** | SylvieApp | Idea counter reset | Refresh the browser after collecting ideas | Collected ideas persist across sessions | Idea count resets to zero; unlocked items disappear | P3 | Store progress in local storage or prompt about progress |
| **BUG-6** | Sleepy | Dim lights flicker | Click “Dim lights” then navigate between categories | Brightness remains dim | Brightness flickers back to full and resets | P2 | Persist dim state across route changes; apply CSS correctly |
| **BUG-7** | Sleepy | Offline fallback | Disconnect network; open a lullaby video | App shows an offline message or cached audio | Broken video thumbnail; nothing plays | P1 | Provide offline fallback content or clear messaging |
| **BUG-8** | EliasApp2 | Tap target overlap | In Construction scene on phone, tap near two overlapping cranes | Only one crane moves | Both cranes activate or wrong element moves | P2 | Increase spacing; enlarge hit areas |
| **BUG-9** | EliasApp2 | Sound toggle reset | Turn sound off; refresh page | Sound remains off | Sound reverts to on | P3 | Persist sound preference in local storage |

| ID | App | Screen/ feature | Steps to reproduce | Expected result | Actual result | Severity | Recommended fix |
|---|---|---|---|---|---|---|---|
| **BUG-10** | EliasApp2 | Long press selects text | Long press on an interactive object (e.g., truck) | Nothing happens beyond the action | Text selection overlay appears; blocks view | P2 | Disable text selection on interactive elements |

## 6. Evidence-to-Feature Mapping

| App | Feature | Observed evidence | Child need or goal | Classification | Reasoning | Recommendation |
|---|---|---|---|---|---|---|
| **SylviePhonetics** | Generic “Great job!” praise after correct answers | Games provide uniform praise regardless of the child’s response; no hints | Contingent feedback | **Neutral** | Generic praise does not harm, but it fails to teach or scaffold; the child misses an opportunity to reflect | Replace with specific feedback: repeat the target phoneme, provide an example word, invite the child to try again |
| **SylviePhonetics** | Letter selection from pictures | The child chooses between two letters after seeing a picture; sometimes letters appear before sounds | Phonological awareness progression | **Potentially harmful** | Introducing letters too early may cause guessing based on the first letter or picture, undermining sound awareness and causing frustration | Separate sound games from letter games; maintain audio-first learning until phoneme isolation is secure |
| **SylvieApp** | Calm Down page with breathing animation | The app guides slow breathing with phrases such as “You are safe” and references to God | Interoception & emotional regulation | **Direct** | Slow breathing and supportive phrases align with evidence that mindful breathing helps under-5s regulate | Integrate shorter breathing prompts into challenging scenes; ensure phrases are inclusive for broader audiences |

| App | Feature | Observed evidence | Child need or goal | Classification | Reasoning | Recommendation |
|---|---|---|---|---|---|---|
| **SylvieApp** | Idea (star) collection unlocks new items | Collecting stars unlocks additional features after three stars | Perfectionism & motivation | **Potentially harmful** | Extrinsic reward loops can encourage Sylvie to fixate on collecting rather than enjoying play; may trigger frustration when progress resets | Replace with narrative unlocking tied to collaborative achievements rather than star counts |
| **Sleepy** | YouTube video links to Disney songs | The site embeds videos of Aladdin, Frozen, Moana and other licensed content | Sleep routine support | **Harmful** | These videos are long, visually stimulating, and connect to YouTube's algorithm; they can suppress melatonin and delay sleep onset | Remove these links; replace with short, audio-first stories and curated lullabies hosted locally |
| **Sleepy** | Dim lights & Sleepy mode toggles | Buttons that apply warm filter and dark overlay | Sleep routine support | **Direct** | Warm colours and reduced brightness help prepare for sleep; positive design | Ensure mode persists across navigation; automatically enable for all bedtime content |
| **EliasApp2** | Parent co-play overlay at launch | On loading, an overlay invites a caregiver to sit with Elias | Co-regulation & parent involvement | **Direct** | The overlay prompts adult involvement and emphasises co-play; strongly supported by research | Keep overlay but ensure it cannot be dismissed without reading; include age-appropriate instructions |

| App | Feature | Observed evidence | Child need or goal | Classification | Reasoning | Recommendation |
|---|---|---|---|---|---|---|
| **EliasApp2** | Multiple interactive elements in Construction scene | More than five clickable tools on one screen | Attention span limits | **Indirect/ harmful** | Too many interactive elements exceed cognitive load for a toddler, leading to confusion or hyperfocus | Limit to three main actions per screen and guide sequential exploration |
| **EliasApp2** | Screwdriver World warns about real tools | Text at the top warns that electric tools are for adults only | Safety & real-world transfer | **Direct** | Modelling safe tool use and explicit warnings align with safety goals; encourages real-life learning | Retain and expand warnings to other scenes; consider adding safe handling language in audio |

## 7. Prioritised Improvement Roadmap

The table below ranks the most important changes across the suite. Priority is based on child impact (high, medium, low), safety risk, educational value, evidence strength, and development effort.

| Priority | App | Change | Why it matters | Child impact | Evidence strength | Effort | Acceptance crite |
|---|---|---|---|---|---|---|---|
| **1** | Sleepy | Replace YouTube embeds with locally hosted audio-only stories and lullabies; add mandatory “lights-out” mode | Current videos delay melatonin production and risk overstimulation; removing them immediately mitigates a high safety risk | High | A | Medium | All media hosted locally; screen di to black after sta no external recommendatio |

| Priority | App | Change | Why it matters | Child impact | Evidence strength | Effort | Acceptance crite |
|---|---|---|---|---|---|---|---|
| **2** | SylviePhonetics | Implement scaffolded feedback and hint system for wrong answers | Lack of feedback undermines learning and increases frustration; hints support frustration tolerance and flexible thinking | High | A | Medium | Every incorrect answer triggers a hint, then model the phoneme, th invites another attempt |
| **3** | All apps | Add session timers and natural closing rituals aligned with recommended durations (Sylvie: 10–12 min; Elias: 5–8 min) | Research emphasises short sessions with gentle endings to protect attention span and emotional regulation | High | A | Medium | After timer expir app prompts to end and encourages offli play; cannot continue withou caregiver resetti |
| **4** | SylvieApp | Remove or redesign star/idea collection; integrate relational or narrative unlocking | Extrinsic reward loops risk perfectionism and stress; narrative unlocking fosters motivation | Medium | B | Medium | Features unlock through collaborative events; no nume counter displaye |
| **5** | EliasApp2 | Add audio vocabulary narration for each action and limit interactive elements to three per screen | Without language modelling, vocabulary growth depends entirely on parent; too many actions overwhelm toddlers | Medium | B | Low | Each tap triggers an Australian-accen label; no screen >3 interactive objects |

| Priority | App | Change | Why it matters | Child impact | Evidence strength | Effort | Acceptance crite |
|---|---|---|---|---|---|---|---|
| **6** | Sleepy | Shorten content length (<30 min) and embed co-regulation prompts (“snuggle now”, “turn page together”) | Reduces screen exposure near bedtime; invites parent involvement | Medium | B | Low | All items under 30 min; closing ritual invites to read a physical book |
| **7** | SylviePhonetics | Separate sound games from letter games; follow systematic sequence (rhyme → syllable → onset/rime → phoneme isolation → blending → grapheme–phoneme mapping) | Mixing letters too early confuses the child and encourages guessing | Medium | A | Medium | Each mini-game targets only one stage; letters appear only afte phoneme isolati mastery |
| **8** | SylvieApp | Integrate emotional and social narratives into scenes (characters model feelings and repair) | Embeds emotional literacy into play, meeting therapy goals | Medium | B | Medium | At least one scer shows a characte expressing and resolving an emotion |
| **9** | EliasApp2 | Introduce parent gate for category changes and session start | Prevents unsupervised use; ensures co-play | Medium | C | Low | Long press or simple math challenge requir to switch categories or beg play |

| Priority | App | Change | Why it matters | Child impact | Evidence strength | Effort | Acceptance crite |
|---|---|---|---|---|---|---|---|
| **10** | All apps | Add accessibility features (audio labels, high-contrast mode, closed captions) | Improves usability for diverse families; ensures inclusive design | Low | C | Medium | Accessibility settings added; t labels have audi high-contrast mo available |

**Quick wins.** Add audio vocabulary to EliasApp2; persist sound and dim settings in Sleepy; add local storage for idea count in SylvieApp; disable horizontal scrolling on the SylvieApp map; persist sound toggle across sessions; disable text selection on interactive elements in EliasApp2; hide letter labels in early phonics games; apply alt text to all icons.

**Medium changes.** Implement session timers with closing rituals; refactor Sleepy to host audio stories and remove YouTube; add hint and feedback system in SylviePhonetics; redesign star unlocking; integrate emotion modelling into SylvieApp scenes; limit interactive elements in EliasApp2; implement parent gates.

**Strategic changes.** Expand voiceover support across all apps; develop a parent dashboard summarising usage and progress; integrate optional offline printables or physical activities; align SylviePhonetics with the NSW phonics scope & sequence; design cross-app bridging prompts that reference real family experiences (e.g., check the vegetable garden after playing farm). Consider obtaining licences for high-quality children's music rather than using YouTube content.

## 8. Test Cases for the Next Release

### Universal test cases

1. **Session timer**: Launch each app, play continuously until the recommended time (10 min for Sylvie; 5 min for Elias) elapses. Verify that a closing ritual appears and invites the child to finish and show a caregiver. Ensure the app does not progress further until the caregiver confirms.
2. **Parent gate**: Attempt to access settings or start a new game without an adult. Confirm that a gate (e.g., long press or solving a math problem) prevents the child from proceeding independently.
3. **Accessibility**: Toggle high-contrast mode and confirm that text and icons meet WCAG contrast ratios. Use a screen reader to ensure alt text is present and logical.
4. **Offline resilience**: Disconnect the internet while using each app. Confirm that content either plays from local storage (if applicable) or that a clear offline message appears. Ensure no broken thumbnails or unhandled errors.
5. **Multi-device persistence**: Collect progress or settings on one device (e.g., toggling sound, unlocking items) then reopen on another. Confirm that progress persists if it is intended to; ensure privacy if data is device-specific.
6. **Rapid tapping**: Simulate rapid tapping on interactive elements for 30 seconds. Confirm that no game freezes, miscounts or unhandled exceptions occur.
7. **Dim mode**: In Sleepy, turn on dim mode and navigate through categories. Confirm that brightness and warm colour filters persist across pages and remain active until turned off.

8. **Audio narration**: With sound toggled on, tap each interactive element in the updated apps. Confirm that an Australian-accented voice names the object or action. With sound off, verify silence.
9. **Breathing break**: Play SylviePhonetics or EliasApp2 until a movement/breathing break appears. Ensure the child must interact with the break (e.g., stand up, breathe) before continuing.
10. **Accessibility of parent prompts**: Check that prompts encouraging co-play are visible, clear and not hidden behind small toggles. Ensure parents can access instructions easily.

## SylviePhonetics test cases

1. **Hint sequence**: Answer incorrectly; verify that a hint appears ("Listen to the first sound again") followed by modelling and a second chance. Ensure the child cannot progress without hearing the hint.
2. **Sequence separation**: Play rhyme, syllable and phoneme games. Confirm that no letters appear in rhyme or syllable games. Ensure grapheme mapping appears only after phoneme isolation levels are completed.
3. **Movement break**: After five rounds, verify that a movement prompt appears ("Jump like a kangaroo!") and that the game pauses until the child resumes.
4. **Parent prompt**: In parent assist mode, check that each game includes at least one prompt inviting a caregiver's involvement (e.g., "Ask Dad to clap with you").
5. **Progress tracking**: After completing a set of games, refresh the page. Confirm that progress (e.g., completed levels or badges) persists if appropriate, or resets if designed to clear after each session. Ensure there is an option to reset progress manually via the parent gate.

## SylvieApp test cases

1. **Emotion modelling**: Enter a scene with the new emotion narrative (e.g., farm scene where a pig knocks over blocks). Confirm that the character expresses an emotion, names it and models repair; ensure the child can participate in the repair.
2. **Star removal**: Play several scenes and confirm that unlocking new features is based on narrative or joint achievements rather than star counts. Ensure there is no numeric counter.
3. **Closing ritual**: After 10 minutes, verify that the app transitions to a gentle ending and invites the child to continue play offline (e.g., make a real fairy garden). Ensure the child cannot continue without restarting.
4. **Audio narration of labels**: Tap labels and icons; verify that an Australian-accented voice reads them aloud. Ensure the voice is clear and there is an option to mute narration.
5. **Map stability**: On tablet and phone, drag the world map in all directions. Confirm that horizontal scrolling is disabled and icons remain aligned.

## Sleepy test cases

1. **Media hosting**: Attempt to play each lullaby and story; verify that no YouTube or external link loads. Confirm that all media plays locally and offline.
2. **Lights-out mode**: Start a Sleepy session; ensure the screen fades to black within 30 seconds of starting and only audio continues. Verify that tapping brings back visuals only with parent confirmation.
3. **Content length**: Check that each item is under 30 minutes. Confirm there are no loops or playlists that exceed the limit. After finishing, verify that a closing ritual invites the parent to continue offline bedtime routines.
4. **Age-appropriate warning**: Launch the app with an age setting under two; verify that a warning appears discouraging use and recommending no screen time for that age..

5. **Dim mode persistence**: Turn on dim mode; navigate across categories; ensure that the warm filter remains active and brightness stays low until manually turned off.

### EliasApp2 test cases

1. **Vocabulary narration**: Tap each object; confirm that an Australian-accented voice names the object and action. With sound off, ensure there is no narration.
2. **Interactive elements limit**: Verify that no scene has more than three interactive elements. Attempt to interact with hidden elements; ensure they remain inaccessible until sequentially unlocked.
3. **Session timer**: Play continuously; check that after 5–8 minutes the app presents a closing ritual encouraging the child to play with real trucks and prevents further digital play until reset by a caregiver.
4. **Parent gate**: Attempt to switch categories or start a game; ensure a long press or math question appears. Confirm that a toddler cannot bypass the gate accidentally.
5. **Co-play prompts**: Read prompts in each scene; confirm that at least one prompt encourages the caregiver to narrate or imitate (e.g., "Make the truck sound together"). Ensure prompts are visible and not hidden behind menus.

## 9. Questions for Parent Observation

1. **Sylvie's regulation**: When Sylvie makes a mistake in SylviePhonetics, does she remain calm and attempt again, or does she show signs of frustration or shutting down? Does the new hint system help her persist?
2. **Engagement and transfer**: After playing a scene in SylvieApp, does Sylvie spontaneously continue the theme in real-world play (e.g., building a fairy garden outside, telling a story)? Does she talk about the characters' feelings or actions?
3. **Sleepy routine**: Does Sleepy's lights-out mode help Sylvie wind down, or does she become overstimulated by the screen? Do bedtime struggles decrease when using audio-only stories?
4. **Elias's shared attention**: Does Elias look toward you and share attention when using EliasApp2? Does he repeat the vocabulary words? Does he return to his physical vehicles and tools after the session?
5. **Co-play participation**: Are parent prompts effective in drawing you into the play? Do you find them intrusive or helpful? Do the children invite you without prompts?

## 10. Final Recommendation

For the next development sprint, focus on **remediating Sleepy** and **strengthening feedback and session design across all apps**. Sleepy poses the most significant safety and developmental risks due to bright screens, long videos and external links; replace YouTube content with locally hosted audio stories, enforce lights-out mode, and add session timers. In parallel, implement scaffolded hints in SylviePhonetics and restructure its sequence to separate sound from letter tasks. Remove extrinsic star collection in SylvieApp and integrate emotional narratives and closing rituals. For EliasApp2, introduce audio vocabulary narration and limit interactive elements to avoid overload. Success will be evident when sessions naturally end within recommended durations, children remain regulated, caregivers are consistently involved, and the apps guide the child toward real-world play. Continuous observation by Josh and Kristy will ensure iterative refinement and alignment with therapy goals.

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[1] Sylvie's Phonics Adventure
https://joshualparris.github.io/SylviePhonetics/